PRE-BOOTCAMP READINESS SESSION

AI in Biomed Boot Camp

Getting everyone set up on AWS — GW Research Gateway

60 minutes · Rocky Linux 8 Remote Desktop · Jupyter + Ollama

WHY RESEARCH GATEWAY

Why this bootcamp is using Research Gateway

Why the environment changed

- Pegasus storage is unavailable for this bootcamp window.
- Research Gateway gives everyone a working Linux environment now.
- That keeps the schedule intact and preserves the hands-on session.

What still transfers

- ✓ Jupyter notebooks
- ✓ PyTorch / Torch
- ✓ Ollama + local models
- ✓ bash / CLI workflows
- ✓ Python environments (venv, pip)

The main thing that does not transfer here is Slurm job scheduling.

AGENDA

What we'll cover in this hour

We start the cloud desktop first, then use the provisioning window well.

0:00–0:05	**Launch the desktop**	Sign in, join the project, and start the Rocky Linux desktop
0:05–0:20	**GW computing + Bash basics**	Research Gateway, Pegasus, cloud vs. on-prem, and the shell commands we will use today
0:20–0:35	**Open Jupyter**	Log into the desktop, create the venv, and launch Jupyter
0:35–0:50	**Run the Ollama notebook**	Open ollama_aws.ipynb, start the model download, and confirm it responds
0:50–1:00	**Readiness check + help**	Troubleshooting, support, and the five things that must be green

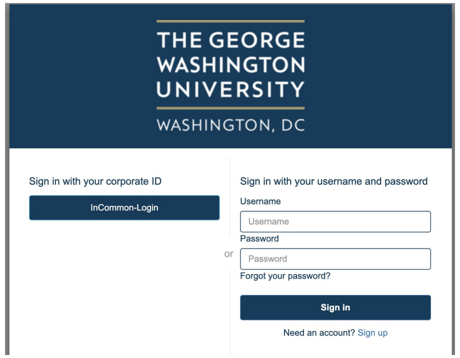
Start your cloud desktop right away

0:00–0:05

Launch first so the environment is ready by the time we reach the hands-on work.

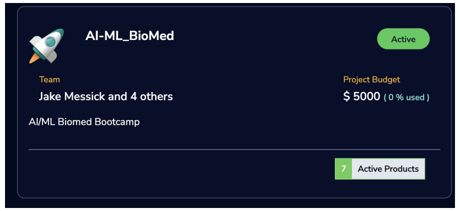
1 Log in

rg.arc.gwu.edu ⟶ InCommon ⟶ Sign in with GW



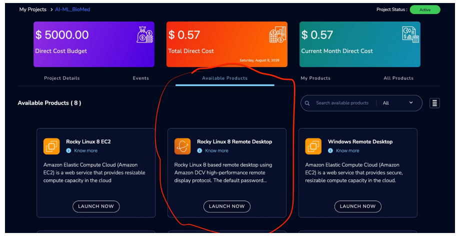
2 Open the project

Select AI-ML_BioMed



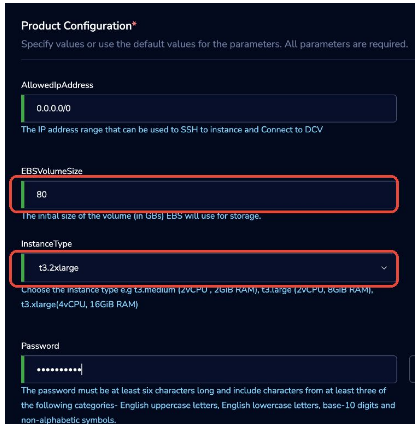
3 Choose the desktop

Rocky Linux 8 Remote Desktop



4 Configure + launch

80 GB · t3.2xlarge · Launch Now



Provisioning takes about 15 minutes, so start it before the rest of the session begins.

BEFORE THE SESSION

Complete this access check before bootcamp day

If your first login stops at "requires admin approval," get that cleared before the session starts.

1. **Sign in once ahead of time**

   Do not make your first Research Gateway login at the start of the bootcamp.

2. **Confirm the project and product**

   You should be able to open AI-ML_BioMed and see Rocky Linux 8 Remote Desktop.

3. **Resolve blockers before the session**

   If approval is pending or the project is missing, contact RTShelp before bootcamp day.

What to know while the desktop provisions

0:05–0:20

1 Cloud vs. on-prem at GW

Why this bootcamp uses Research Gateway now, and when Pegasus or Raptor are the better fit for real workloads.

2 How access works

You sign in through InCommon, launch inside your project, and need to keep project budgets in mind.

3 Linux / Bash basics

Paths, files, history, tab completion, and the commands that keep people unstuck.

4 Notebooks & Google Colab

Jupyter is the working environment for this bootcamp. Colab is a useful comparison point and fallback.

CLOUD VS. ON-PREM

Cloud and on-prem solve different problems

Cloud (Research Gateway / AWS)

- Runs as an individual desktop or VM
- Fast to provision for teaching and short projects
- Good for notebooks, package installs, and local models
- Stop the instance when you are done

On-prem (Pegasus / Raptor)

- Shared institutional compute and storage
- Better for scheduled HPC / HTC workloads
- Built around queues, allocations, and shared filesystems
- Best when you need scale, throughput, or longer-running workflows

Today we use cloud for fast, uniform setup; many production research workloads still belong on GW's on-prem systems.

LINUX / BASH

Shell commands everyone should see today

Navigation and files

- pwd and ls to see where you are
- cd to move through directories
- mkdir, cp, mv, and rm for basic file work
- cat or less to inspect text files
- nano or vi to edit files

Habits that keep people unstuck

- Tab completion to avoid typing mistakes
- Up-arrow history to rerun and edit commands
- man or --help to check syntax quickly
- source my_new_env/bin/activate to enter the Python environment

The goal is not full Linux fluency; it is enough command-line comfort to navigate, activate the environment, and recover from simple mistakes.

Google Colab vs. Jupyter on your cloud desktop

Google Colab

- Runs entirely in the browser
- Easy sharing and quick experiments
- Free tiers are convenient but limited
- Sessions expire and the environment is not fully under your control

Jupyter on the cloud desktop

- You control packages, storage, and the runtime
- Persistent project workspace
- Closer to real research-system workflows
- The same environment you will use in the bootcamp

Colab is great for trying an idea; the cloud desktop is where this bootcamp actually runs.

Open Jupyter on the cloud desktop

0:20–0:35

Log into the desktop

- A new browser window opens with your desktop.
- Log in with the password you created at launch.
- If the password box is hidden, click inside the window and drag upward.

Then, in a terminal

```
sudo dnf install python3.12 -y
python3.12 -m venv my_new_env
source my_new_env/bin/activate
pip install notebook
jupyter notebook
```

Open the notebook and start the download

Once Jupyter opens, open ollama_aws.ipynb in ProjectStorage and run the model-pull cells first.

The download can run in the background while you keep moving through the notebook.

Notebook:

```
ProjectStorage/ollama_aws.ipynb
```

How notebooks and Ollama fit together

0:35–0:50

Notebooks

- Use the notebook to run code, inspect outputs, and keep the work in one place.
- The shell still matters for setup and troubleshooting.
- This is the interface you will use throughout the bootcamp.

Ollama and models

- Ollama lets the notebook talk to a local model.
- Model choice is a tradeoff among quality, speed, and memory.
- By the end of this section, the model should be pulled and responding.

If something breaks, start here

0:50–1:00

Login fails

Check your InCommon/GW sign-in first; a first login may still need admin approval.

Instance won't launch

Check AI-ML_BioMed access, project budget, and the 80 GB / t3.2xlarge settings.

Jupyter won't start

Activate the venv first (source my_new_env/bin/activate), then try again.

Package or model missing

Re-run the notebook setup or model-pull steps and confirm the venv is active.

Where to get help

Use RTShelp or office hours before the bootcamp if you cannot complete the setup.

Watch the budget

Stop the instance when you are done so the shared project budget is not consumed unnecessarily.

READINESS CHECK

You're ready when…

- ✓ Logged into Research Gateway (rg.arc.gwu.edu)
- ✓ Launched Rocky Linux 8 Remote Desktop (80 GB · t3.2xlarge)
- ✓ Jupyter running from your venv
- ✓ Ran ollama_aws.ipynb end to end
- ✓ Model pulled and confirmed responding in a notebook

Come to the bootcamp with all five green.

APPENDIX

Step-by-step setup reference

Screenshots, setup guide, and notebooks are in the BootcampSquared GitHub repo.
Scan the QR code to open it.

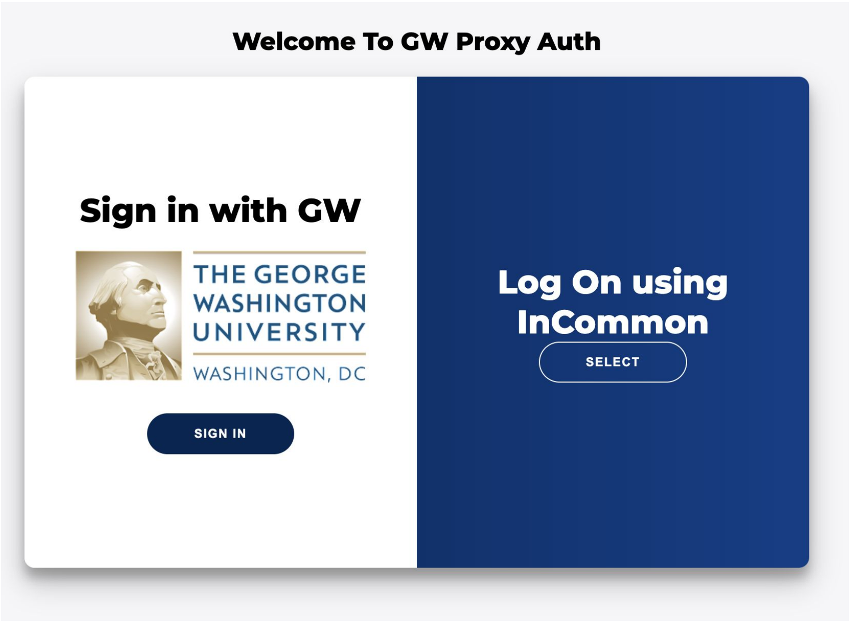
Sign in to Research Gateway

1 Go to rg.arc.gwu.edu

Choose InCommon-Login.

2 Sign in with GW

A first login may show “requires admin approval.” That is normal; do this step before the session if you can.



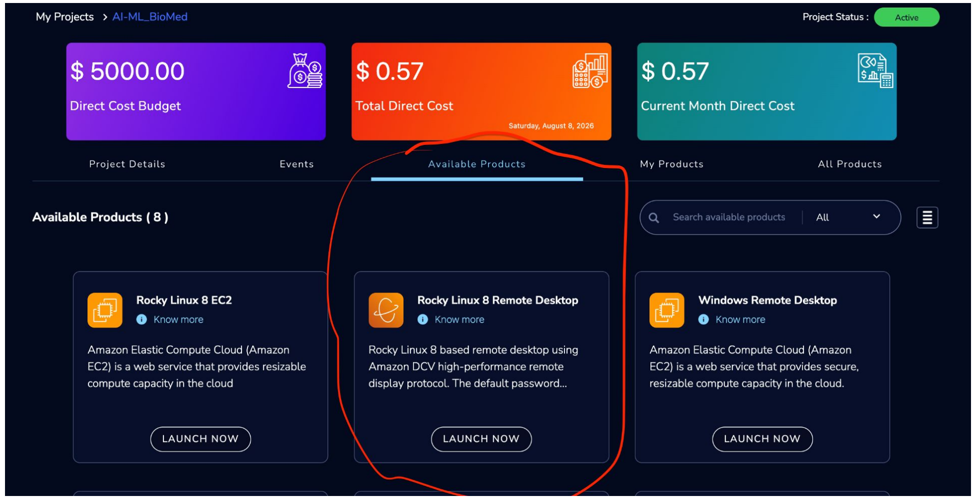
Open the project and choose the desktop

3 Open AI-ML_BioMed

From My Projects, open the AI-ML_BioMed project card.

4 Available Products → Rocky Linux 8 Remote Desktop

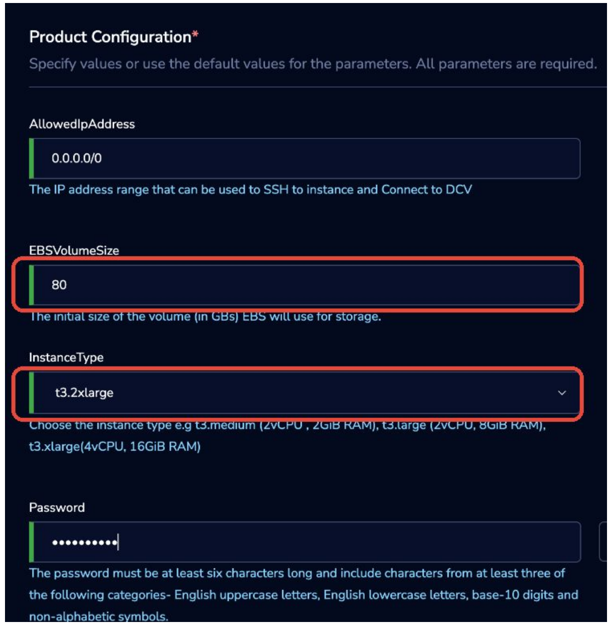
Under Available Products, choose Rocky Linux 8 Remote Desktop.



Configure, launch, and sign in

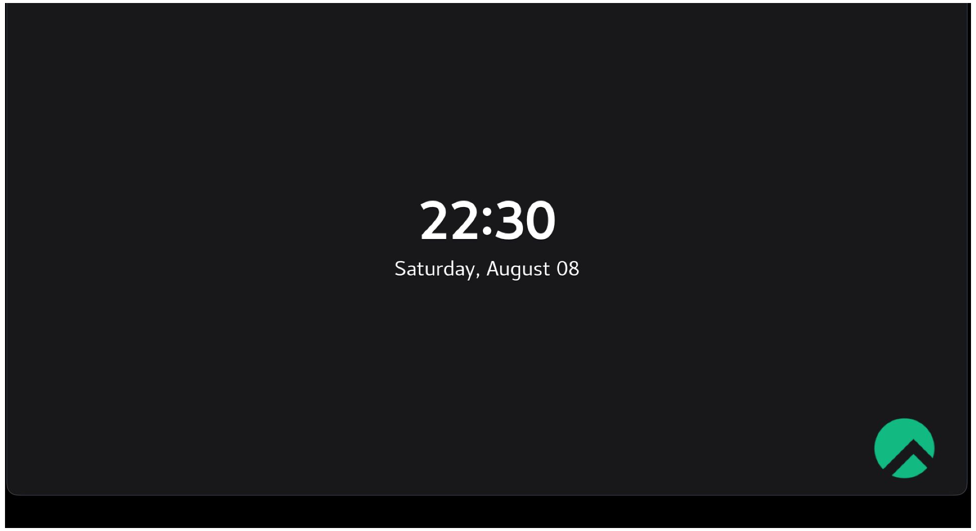
5 Set storage, instance type, password

Set EBSVolumeSize to 80, InstanceType to t3.2xlarge, choose a password, and click Launch Now.



6 Log into the desktop

A new browser tab opens with the desktop. Log in with your password; drag upward if the password box is hidden.



Open Jupyter and start the notebook

Open a terminal on the desktop and run:

```
sudo dnf install python3.12 -y
python3.12 -m venv my_new_env
source my_new_env/bin/activate
pip install notebook
jupyter notebook
```

Then open:

`ProjectStorage/ollama_aws.ipynb`

Run the model-pull cells first so the download finishes while you keep working.

Jupyter opens in a browser on the desktop. The virtual environment must be active before you launch it.